24CSA301-Machine Learning

Lecture: Computational Learning Theory

Overview

- Sample complexity
- ε- exhausted version space
- PAC learning
- agnostic learner
- Vapnik–Chervonenkis (VC) dimension

Computational Learning Theory

What general laws constrain inductive learning?

Inductive learning: learning general rule from specific training sample

We seek theory to relate:

- Probability of successful learning
- Number of training examples
- Complexity of hypothesis space
- Accuracy to which target concept is approximated
- Manner in which training examples presented

Sample Complexity

- How many training examples are sufficient to learn the target concept?

Sample Complexity = Number of training samples needed to achieve good performance

- **It is important because:**
- If too few samples are used:
  - Underfitting occurs.
  - The model cannot learn patterns.
- If sufficient samples are available:
  - Better generalization.
  - Higher prediction accuracy.

- Sample complexity increases when:
  1. Higher accuracy is required
  2. More confidence is desired
  3. The hypothesis space becomes larger

ε- exhausted version space

- Version Space is: the set of all hypotheses that are consistent with the training data.
- ε : represents maximum acceptable error

- ε- exhausted: Consider a hypothesis space H, target concept c, instance distribution X, and set of training examples D of c. The version space $VS_{H,D}$,is said to be ε- exhausted with respect to c and D, if every hypothesis h in $VS_{H,D}$ has error less than ε with respect to c and D.

PAC Learning

- Probably Approximately Correct Learning is a framework for mathematical analysis of Machine Learning
- Key parameters:

ε (epsilon)

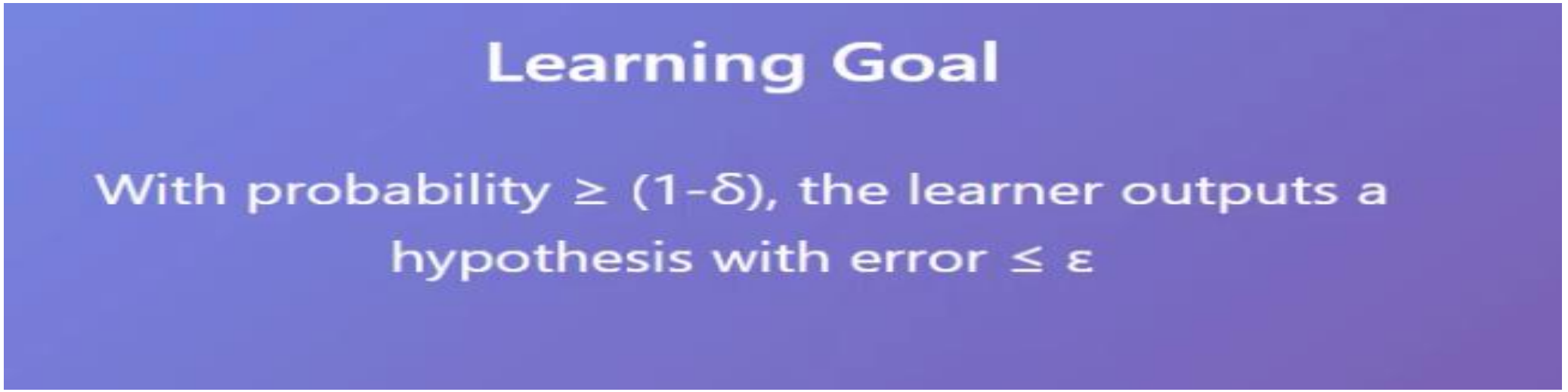
Error tolerance

δ (delta)

Confidence parameter

m

Sample size



- Goal: With high probability, the selected hypothesis will have low error
- Assume there is no error in the data

Agnostic Learner

- An **agnostic learner** is a learning algorithm that finds a hypothesis whose error is **as close as possible to the minimum achievable error**, even when no hypothesis in the hypothesis space can perfectly classify the training data.

Vapnik–Chervonenkis (VC) dimension

- The VC dimension is a measure of the complexity of a model
- The VC dimension provides both upper and lower bounds on the number of training examples required to achieve a given level of accuracy.
- The **VC (Vapnik–Chervonenkis) Dimension** is a measure of the **learning capacity or complexity of a machine learning model**. It tells us **how powerful a model is in fitting different patterns in data**.

VC Dimension and Model Complexity

- The VC dimension is directly tied to a model's complexity:
- Higher VC Dimension: Indicates a more complex model capable of learning intricate patterns.
- Lower VC Dimension: Suggests a simpler model with limited learning capacity.

Balance Between Complexity and Generalization:

- Overfitting: A model with a very high VC dimension may overfit, memorizing the training data instead of generalizing.
- Underfitting: A model with a very low VC dimension may underfit, failing to capture the patterns in data.